Quadcopter Battery Mounting System

Mechanical Design and Interface

System Specifications

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Version 1

VERSION HISTORY				
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1 PURPOSE

The purpose of this system concept is to design and define a mechanical structure that houses a battery in a way that reduces swap time and operator error. The structure must allow quick, repeatable battery changes while maintaining centre of gravity, structural integrity, and clean routing of data/power wiring. It should enable reliable swaps without complex machines such as drone battery-swapping systems, docking stations, or even hand tools.

The goal is a simple, low-cost design that can be 3D-printed or built with composite materials and carbon-fibre reinforcement, integrated into the airframe, and mass-produced. The design thinking focuses on manufacturability, minimal component count, subsystem integration, low cost, safety/failure considerations, and scalability.

2 OVERALL TECHNICAL APPROACH

2.1 Product Context and Related User Documentation

The product is a battery mounting system for a 225 mm FPV quadcopter (5" class) with total aircraft mass between 250 g and 1 kg. It replaces traditional strap-based LiPo mounting with a tool-less, slide-in, snap-lock, quick-swap battery system using an integrated connector and carbon-fiber composite casing. All documentation references standard FPV airframe layouts, Part 107 operational constraints, and the intent of MIL-STD-810 / MIL-STD-882E safety practices.

2.2 Data Requirements

1. Battery geometry and mass (6S LiPo, FPV-class sizes).
2. Drone CG envelope (±3 mm longitudinal, ±2 mm vertical).
3. Vibration spectrum for 5" quads (20–300 Hz typical).
4. Crash load estimates (30–50 g equivalent, depending on impact scenario).
5. Electrical current draw profile of ESC/power bus.
6. Swap-cycle durability data (500–1000 cycles).
7. Future optional: telemetry data for BMS-based failure prediction.

2.3 Design and Implementation Constraints

1. Total mount + casing added mass ≤3 g beyond the bare battery.
2. Must support <10 s battery swap.
3. Must withstand operational vibration and crash loads without detaching.
4. Must not violate drone CG envelope.
5. No tools; must use interlocking, self-aligning mechanism.
6. Must allow automatic electrical connector engagement (no manual plugging).
7. Thermally neutral: must not trap heat or increase fire risk.
8. Manufacturing must be viable in CF composite with small hardware components.
9. Must comply with Part 107 operational conditions.

3 FUNCTIONAL REQUIREMENTS

3.1 Capabilities Supported

The system supports fully tool-less battery insertion and removal with a slide-in mechanism that automatically engages the main power connector upon full seating. A positive snap-lock provides crash-resistant retention while maintaining highly repeatable CG positioning between swaps. The design accommodates standard 6S LiPo batteries within a protective carbon-fibre casing and is engineered to survive 20–300 Hz vibration, high-g manoeuvres, and typical quadcopter crash loads. It enables fast battery changes in under 10 seconds, maintains structural integrity under pull-tests and impact scenarios, and is architected to remain compatible with future BMS telemetry integration.

3.2 Capabilities Not Supported

The system does not include environmental testing beyond mechanical vibration and crash scenarios, and it does not provide real-time battery telemetry in this version. No active thermal regulation or onboard charging is provided, and the design does not support multiple battery sizes unless specifically adapted in future revisions.

3.3 Performance Targets

The system is required to enable battery swaps in under 10 seconds and maintain CG consistency within ±3 mm longitudinally and ±2 mm vertically. The retention mechanism must withstand at least 30–50 g equivalent loads and demonstrate a quick-swap durability of at least 500 cycles, with a stretch goal of 1000 cycles. During vibration testing, the battery must not detach or partially unlock, and the electrical connector must remain fully stable with no intermittency under vibration or repeated cycling.

4 DESIGN STATEMENT

The battery system is a carbon-fibre-reinforced slide-in mount with a rear-locking snap mechanism and integrated high-current electrical connector.

The user inserts the battery from the top, the casing self-aligns, the connector auto-mates, and the snap-lock secures the pack. Crash resilience, vibration tolerance, CG stability, and tool-less operation drives all design choices.

The architecture enables safe integration into FPV quads while eliminating straps and reducing operational errors.

- **Inputs**

Variable	type	Description
Battery Mass	Float	Mass of 6S LiPo; used for CG and load calculations.
Battery Dimensions	Vector	Defines casing and clearance geometry.
Frame Geometry	CAD/Parametric	Defines mounting rails and integration points.
Vibration Profile	Spectrum	Determines structural and locking requirements.
Crash Load Estimate	Float	Used to size retention and snap-lock strength.
Electrical Current Draw	Float	Determines connector specification.

- **Outputs**

Variable	type	Description
Mount Geometry	CAD Model	Defines final mount and casing.
Load Ratings	Numbers	Pull-test, vibration, crash survivability specs.
CG Envelope Compliance	Bool/Report	Pass/fail for CG repeatability.
Swap-Cycle Durability	Cycles	Verified lifetime of mechanism.
Safety Analysis Results	FMEA/FTA Docs	Identified risks and mitigations.

5 COMPUTER RESOURCE REQUIREMENTS

5.1 Resources

1. CAD tools (Fusion360, SolidWorks, or equivalent).
2. FEA tools (optional) for load estimation.
3. Spreadsheet/FMEA tools for safety analysis.
4. No onboard computational load (mount is mechanical).

5.2 Previously Developed requirements

N/A.

5.3 New Requirements

1. Support for automated connector alignment geometry.
2. Reliable snap-lock design with measurable retention force.
3. CF composite manufacturability constraints.

6 MANAGEMENT PLAN

6.1 Schedule and Milestones

Starting with concept using CAD based on system parameters. Followed by Structural verification (FEA) , 3D printed prototype fabrication , Detachment tests based on FEA, Iterations and final validations.

6.2 Risk Assessment

DFMEA, effects on CG due to static/dynamic conditions, battery detachment during crash, connector misalignment , structural failure and manufacturability issues.

6.3 Configuration Management Items

1. CAD files (mount + casing).
2. Test procedures and results.
3. FMEA/FTA documents.
4. Material specifications.
5. Revision-controlled connector geometry.

7 REFERENCES

[1] https://arxiv.org/pdf/2511.13122

[2] https://prism.sustainability-directory.com/scenario/sustainable-drone-battery-design-for-extreme-environments/

[3] https://www.mdpi.com/2504-446X/7/4/234

[4] https://www.amazon.com/Rhinovo-Lanyard-Controllers-Extended-Adjustable/dp/B0FN43SR2F/ref=sr_1_4_sspa?crid=YSAENNX474JH&dib=eyJ2IjoiMSJ9.X8fnP_jabd_1IHIhBaez5a2lQsJa3JtuDzuFQ0fPWtwkBBt8eoG7t4yLqZDoFfsB26hQoNuv8mKtLT3YhAwfpisuyHo7pt0R2RY-Im5Xn71s-jggTYKB3xdZxDiMcEuc0fN6Fbpooj2cAG0HOWT3goAVZucwffxF3ZHoNCVUqZwpPNQWZYLkqraiy9yFOHXJAWyhR9-uwxrLIZ0g756wj8ZdsoBlTyUsEYYE8igABojeldDdV86sm8X-H-R-87ulVgPNUfaHhCcy0sGRzm-qX1fHa5eizAHvJmMt3LsaaGU.xwTD9pNB5QWwqyREdDBOJXUtJiZHeZnlYbZ06reCfhw&dib_tag=se&keywords=straps+for+drone&qid=1764888534&sprefix=straps+for+drone%2Caps%2C129&sr=8-4-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1

[5] https://www.amazon.com/FPVDrone-Racing-Quadcopter-Freestyle-Battery/dp/B082SPT5D9/ref=sr_1_2?crid=WFN1IHYSKV41&dib=eyJ2IjoiMSJ9.IALCSb9rnr6uegqKKFanwZ9Dqpg6Wv5tqZ9Fb1ZoPX1Wp8sfA24F1z0w619tsNdygQHhq-mvaOF43cuqMaGTM5afzJTEeY5ZUVzQDzz7_zdeYZoj3YUNBaQEsmsVuoU4D8qsVE0fgh1u8EA0UieNrTar1rxTK28JsDXo4TSJQOCO0BLjJYNPMNfuN7d15Zs-Oe9pJtxL9yLlK8kQIi0HTYC0Naphkx8fJvv3vtPQRDv9xvfCdTiw4vqbqZyTvjOvN6bDtpP8wg08KVyKXygxVe4vJgbsLEm5zSDRTIKmhss.zXtOgwO0cP0BqfadUxTZJkEcQrJQ1TrfMlnrvismcn0&dib_tag=se&keywords=drone+chassis&qid=1764888632&refinements=p_72%3A1248963011&rnid=1248961011&sprefix=drone+chassis%2Caps%2C116&sr=8-2